Osmond: A Feature-Rich PCB Designing Tool

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The most important part of embedded system design is the development of printed circuit boards (PCBs). PCB designing has always faced constraints in the development of an efficient design. A lot of design software is available in the market but there are just a handful with engineering excellence and the freedom to customise. This article focuses on Osmond, which banishes all restrictions and artificial limits, and paves new ways to design a PCB.

OsmondCocoa is a simple Macintosh app that helps design PCBs before starting to build the actual circuit board. It is designed using the latest Cocoa application framework. Algorithms and methodologies are enriched with design research from over a decade, whereas the user interface is maintained as modern and sophisticated. It is compatible with Intel and Power PC machines running MacOS X 10.5 or later.

Combining freedom and flexibility

In Osmond, it is very easy to get things on the same grid; all one has to do is type the 'i' key in Osmond, then click at the required new origin. By setting the x-y grid size, parts can be aligned on the new grid.

In terms of precision, Osmond provides a spatial resolution of ten nanometres (0.00001mm), thereby helping design highly-precise PCBs. Here, there is the liberty to design pads of different shapes like circular, rectangular or oval. Also, the parts can be placed anywhere on the board in any orientation.

This extent of design freedom often gives rise to errors, which are minimised by automatic clearance checking, and manual and semi-automatic routing features, thus making it an efficient software.

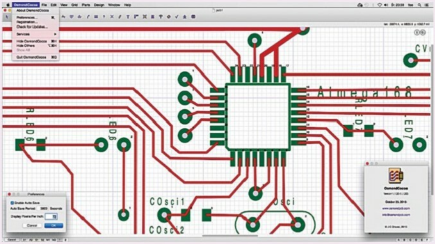

Fig. 1: OsmondCocoa

Inside Osmond

Osmond lets you design a PCB structure meeting specific design rules. It has a set of tools that keep the design error-free. Once a design is complete, standard output like Gerber (RS-274X) or Excellon drill files can be produced, which can be forwarded to PCB fabricators. If you desire more control over the fabrication process, there are options provided to define own panels containing either multiple copies of the one design or several different designs.

Segregating the functionalities

The newest update OsmondCocoa 1.1.33 fixes the bug that restricted rotated pads in Gerber solder paste files render incorrectly. Osmond is free for simple designs and is fully functional, and allows you to design boards up to 700 pins without restriction. For designs over 700 pins, you will need to purchase a licence to print or to output Gerber or Postscript files.

Overview

- **Latest release:** OsmondCocoa 1.1.33
- **Tool for:** Designing PCBs
- **Cost:** Free for small designs
- **Board options:** Virtually unlimited board shapes, sizes and layers
- **Units:** Metric and imperial
- **Scripting support:** Lua
- **Output:** Gerber(RS-274X), DXF, Drill file (Excellon) or Encapsulated PostScript
- **Resolution:** 10nm

About units and layers. When it comes to PCB designing, designers shuttle between metric and imperial units depending on the particular design or application. Osmond provides both of these units and also helps in switching seemlessly between these two units according to the need. It is also equipped with two silkscreen layers, two solder mask layers and two auxiliary layers in addition to signal layers. These can be used to